

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MTS 1C 01—MATHEMATICS—I

(2019—2020 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

Answer any number of questions.

Each question carries 2 marks.

Maximum 20 marks.

1. Find the derivative of $f(x) = x^2 - x$ at $x = 2$.
2. Find $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}$.
3. Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.
4. Find the derivative of $y = (x^2 + 1)(x^3 + 3)$.
5. Give the parameterization of the circle $x^2 + y^2 = 1$.
6. Find $\lim_{y \rightarrow 1} \sec(y \sec^2 y - \tan^2 y - 1)$.
7. Suppose that $F'(x) = x$ for all x and that $F(3) = 2$. What is $F(x)$?
8. Suppose that f is differentiable on the whole real line and that $f'(x)$ is constant. Prove that f is linear.
9. Prove that for the curve $y = c \sin \frac{x}{a}$, every point at which it meets the x -axis is a point of inflection.

Turn over

10. Find the maximum and minimum points and values for the function $f(x) = (x^2 - 8x + 12)^4$ on the interval $[-10, 10]$.
11. Find $\sum_{k=1}^7 (3 - k^2)$.
12. Find $\int_0^1 \frac{(3x^2 + x^4)}{(1 + x^2)^2} dx$.

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 30 marks.

13. If $\sqrt{5 - 2x^2} \leq f(x) \leq \sqrt{5 - x^2}$ for $-1 \leq x \leq 1$, find $\lim_{x \rightarrow 0} f(x)$.
14. Find the linearization of $f(x) = \sqrt{x+1} + \sin x$ at $x = 0$. How is it related to the individual linearizations for $\sqrt{x+1}$ and $\sin x$?
15. An oil slick has area $y = 30x^3 + 100x$ square meters x minutes after a tanker explosion. Find the average rate of change in area with respect to time during the period from $x = 2$ to $x = 3$ from $x = 2$ to $x = 2.1$. What is the instantaneous rate of change of area with respect to time at $x = 2$?
16. Use implicit differentiation to find dy/dx if $6y^2 + \cos y = x^2$.
17. Prove that the curve $y = \frac{x}{1+x^2}$ has three points of inflection and they are collinear.

18. Evaluate $\lim_{x \rightarrow \infty} \frac{x^n}{e^x}$, where n is natural number.
19. Find the area of the region enclosed by the curves $x + y^2 = 3$ and $4x + y^2 = 0$.

Section C

*Answer any one question.
The question carries 10 marks.
Maximum 10 marks.*

20. (a) State and prove the quotient rule of differentiation for positive integers.
- (b) Prove that $\int x^n dx = \frac{x^{n+1}}{n+1} + C, (n \neq -1)$.
- (c) A curved wedge is cut from a cylinder of radius 3 by two planes. One plane is perpendicular to the axis of the cylinder. The second plane crosses the first plane at 45° angle at the center of the cylinder. Find the volume of the wedge.
21. (a) On what interval is $f(x) = x^3 - 2x + 6$ increasing or decreasing?
- (b) Find the asymptotes of the graph of $f(x) = -\frac{8}{x^2 - 4}$.
- (c) Find the equation of the line tangent to the parametric curve given by the equations $x = (1 + t^3)^4 + t^2, y = t^5 + t^2 + 2$ at $t = 1$.

(1 × 10 = 10 marks)

Section B

Answer atleast five questions.

Each question carries 5 marks.

All questions can be attended.

Overall ceiling 25.

13. With the help of suitable diagrams explain demand schedule and demand function.
14. Briefly explain cross price elasticity.
15. Explain the various assumptions on the problem of cost production.
16. Distinguish between short run and long run cost functions.
17. What are the criticism against utility approach ?
18. Find the slope of the average cost curve in terms of average cost and marginal cost.
19. Suppose the price p and quantity q of a commodity are related by the equation $q = 30 - 4p - p^2$. Find elasticity of demand at $p = 2$.

(5 × 5 = 25 marks)

Section C

Answer any one question.

Each question carries 11 marks.

20. (a) Explain the important determinants of price elasticity.
(b) Prove that the elasticity of demand at different points on the same demand curve is different.
21. Use Lagrange multiplier method to optimize $z = 4x^2 - 2xy + 6y^2$ subject to the constraint $x + y = 72$. Also estimate the effect on the value of the objective function from 1-unit change in the constant of the constraint.

(1 × 11 = 11 marks)

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Section A

*Answer at least **eight** questions.*

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

1. Calculate the slope of the tangent line to the graph of $f(x) = x^2 + 1$ when $x = -1$.

2. Find $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$.

3. Find the derivative of $y = \sqrt{x}$ for $x > 0$.

4. Find $\frac{d}{dx} \left[\cos \left(\sqrt{1 + \cos x} \right) \right]$.

5. Find the linearization of $f(x) = \cos x$ at $x = \pi/2$.

6. Show that there is a number c such that $c^3 - c^2 = 10$.

7. Find $\lim_{t \rightarrow 0} \cos \left(\frac{x}{\sqrt{19 - 3 \sec 2t}} \right)$.

8. Suppose that f is differentiable on the whole real line and that $f'(x)$ is constant. Prove that f is linear.

Turn over

9. Find the critical points of $f(x) = 3x^4 - 8x^3 + 6x^2 - 1$.
10. Find the inflection points of $f(x) = x^2 + (1/x)$.
11. Using limits of Riemann sums, establish the equation $\int_a^b c \, dx = c(b - a)$, where c is a constant.
12. Find $\int_0^2 \left(\frac{t^2}{4} - 7t + 5 \right) dt$.

(8 × 3 = 24 marks)

Section B

Answer at least **five** questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Find $\lim_{h \rightarrow 0} \frac{\sqrt{2+h} - \sqrt{2}}{h}$.
14. Show that the line $y = mx + b$ is its own tangent at any point $(x, mx + b)$ on the line.
15. Assume that oil spilled from a ruptured tanker spreads in a circular pattern whose radius increases at a constant rate of 1 ft/s. How fast is the area of the spill increasing when the radius of the spill is 20 ft?
16. Use implicit differentiation to find d^2y/dx^2 if $5x^3 - 7y^2 = 10$.
17. Find the maximum and minimum points and values for the function $f(x) = (x^2 - 8x + 12)^4$ on the interval $[-10, 10]$.
18. Use l'Hôpital's Rule to find $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$.